



Advanced Data Mining D7043E 17054 HT2026

Edge-Ready ECG Classification using CRISP-DM

Group 19:

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1. Business/Application Understanding

1.1 Application Scenario

What problem is going to be solved.

The aim of this project is to solve the problem of automatic individual heartbeat classification from a continuous ECG signal into four clinically defined heartbeat categories (Normal, Supraventricular ectopic, Ventricular ectopic, Fusion), so that abnormal beats can be flagged. The goal is to develop a compact machine-learning system that satisfies the computational constraints of an Edge-AI device (maximum model size 2MB) while achieving good predictive performance on previously unseen ECG recordings.

Automatic ECG classification could be useful in wearable and portable ECG monitoring systems, where continuous ECG signals need to be analysed automatically. These systems could support cardiac monitoring by identifying potentially abnormal heartbeats and generating an alert, reducing the need for manual examination of long ECG recordings.

Running classification directly on an Edge-AI has several advantages: it reduces the amount of data that needs to be transmitted, lowers latency and allows the system to operate autonomously and locally (without no connection to the server). It also increases privacy because raw ECG data remains on the device.

Potential users of such a system are manufacturers of wearable and portable ECG monitoring devices, healthcare professionals who can use it as a supportive tool and researchers.

1.2 Data-Mining Objective

State the machine-learning problem.

Input: An individual heartbeat, represented as a fixed-length segment of ECG signal recording.

Output: One of four class labels for the heartbeat: N (Normal), S (Supraventricular ectopic), V (Ventricular ectopic), F (Fusion).

Type of task: Supervised multiclass classification.

1.3 Project Objectives

List technical objectives.

- Correctly distinguish N/S/V/F heartbeats.
- Maintain robustness to ECG noise.
- Achieve a balance between predictive performance and model complexity.
- Develop a compact machine-learning model that fits within 2 MB size.
- The model must be compatible with MAX78002.

1.4 Success Criteria

Define how to decide whether the project succeeded.

- **Predictive performance:** Macro F1-score ≥ 0.75 .
- **Minority-class performance:** Recall ≥ 0.70 , particularly the minority classes S and F.
- **Noise robustness:** Macro F1-score ≥ 0.70 under moderate noise 6 dB SNR.
- **Model complexity:** Final model size ≤ 2 MB.
- **MAX78002 compatibility:** The model should be accepted by the MAX78002-compatible toolchain and synthesised successfully.

1.5 Constraints

Discuss possible system constraints.

- **Limited Edge-AI memory:** The model must be sufficiently small to meet the memory limitations of the target Edge-AI platform. In particular, the model size is limited to 2 MB, which restricts the number of parameters and the complexity of the architecture.
- **Fixed-point/INT8 computation:** The model needs to support quantised computation, particularly INT8. Quantisation may affect predictive performance and therefore needs to be considered when designing and evaluating the model.
- **Class imbalance:** The heartbeat classes are not equally represented in the dataset. This can make the classifier biased towards the more common classes and make reliable classification of minority classes more difficult.
- **Limited number of subjects:** The dataset contains a limited number of ECG recordings. Therefore, the model must be evaluated carefully to determine whether it generalises to ECG recordings from subjects that were not used during training.
- **Noise:** ECG recordings can contain different types and levels of noise. The model therefore needs to be evaluated for robustness to noisy signals rather than only clean ECG data. It requires testing under clean, moderate, and strong noise conditions.
- **Computational resources:** Training and evaluating different machine-learning models is constrained by the available computational resources. The model has a certain level of complexity.
- **Absence of physical MAX78002 hardware:** The project does not include a physical MAX78002 development kit. Consequently, deployment can only be evaluated at the software/toolchain level, including model quantisation and synthesis. Hardware measurements such as actual inference latency, power consumption, and energy per inference cannot be experimentally verified.

1.6 Risks and Challenges

Identify important risks.

Risk	Why is it important?	Proposed mitigation
Overfitting	ECG signals from the same subject can be similar. A model may learn subject-specific characteristics instead of general patterns, resulting in poor performance on unseen subjects.	Use patient-independent train/test splitting and cross-validation where appropriate. Evaluate the final model only on recordings that were not used during training.
Class imbalance	The majority class (N) may dominate the training process, causing the model to perform well on common heartbeats but poorly on minority classes.	Use appropriate evaluation metrics such as precision, recall and F1-score and not just accuracy
Loss of performance after quantisation	Converting the model from FP32 to INT8 may reduce classification performance or affect some classes more than others.	Compare FP32 and INT8 performance using the same test set and investigate quantisation-aware training if the performance degradation is significant.
High model complexity	A model with too many parameters may achieve good predictive performance but exceed the memory and computational limitations of the Edge-AI platform.	Start with a compact architecture and monitor parameter count and model size throughout development. Investigate simpler architectures where necessary.

2. Data Understanding

2.1 Dataset Description

Database name: MIT-BIH Arrhythmia Database v1.0.0 (PhysioNet)

Number of records: 48

Number of records used: 44 (102, 104, 107, 217 are excluded)

Number of subjects: 47 in total, 43 in the 44 used records

Number of channels: 2

Sampling frequency: 360 Hz

Recording duration: around 30 min each, 1324 min in total

Annotation type: Beat-by-beat, with rhythm and signal quality markers

ECG leads available: MLII, V1 in most records, V2/V4/V5 in a few

4 records (102, 104, 107, 217) have paced beats, the AAMI protocol excludes such records from evaluation, leaving only 44 records.

The 44 records are from 43 subjects, records 201 and 202 are two parts from the same subject. In the DS1/DS2 partition for this project, 201 is in the development set, and 202 is in the final test set, so the partition is record-independent but not subject-independent

2.2 Target Classes

Explain and show any annotations used and excluded.

Final class	Original annotations included	Number of beats	DS1	DS2
N	N, L, R, e, j	90 125	45 866	44 259
S	A, a, J, S	2 781	944	1 837
V	V, E	7 009	3 788	3 221
F	F	803	415	388
Total		10 0718	51 013	49 705

The classes follow the ANSI/AAMI grouping: N has 89.48% of the beats, while F has 0.8%. Class S is 1.85% for DS1 and 3.7% for DS2, so the test set is weighted to the class with the least training data.

Excluded annotations: records 102, 104, 107 and 217 are excluded (Section 2.1), the remaining 44 records have 103724 annotations, of which 100733 are marked as heartbeats.

Class Q (/f,Q): 15 beats remain after the paced records are removed, all of type Q are unclassifiable. They are too few to model and are not part of the four-class task.

Ventricular flutter waves (!): 472 annotations, all in DS1, so treated as non beat markers.

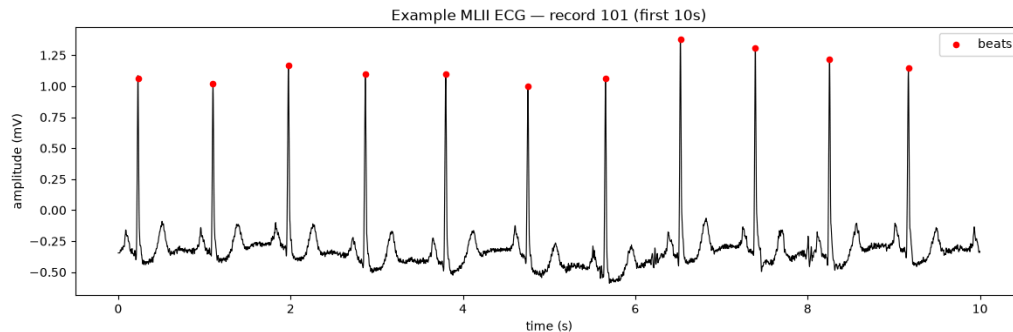
Non-beat annotations: 2519 annotations mark rhythm changes (+), signal quality changes (~), comments ("), non-conducted P waves (x), artefacts (!) and episode boundaries ([,]). None have the heartbeat label.

Count: $100718 + 15 + 2519 + 472 = 103724$.

2.3 Exploratory Data Analysis

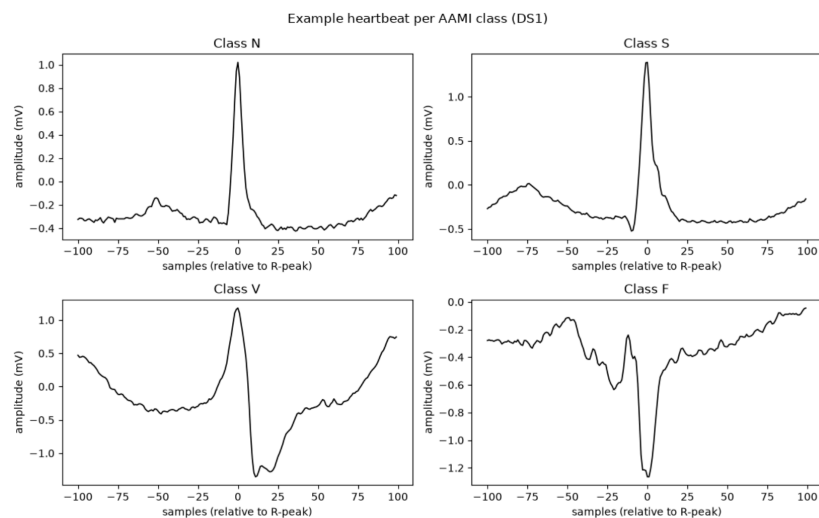
Show and describe findings.

Example normal ECG signal:



The example ECG signal sample of 10 seconds shows a regular sinus rhythm, with each detected beat having a marked R-peak. All QRS complexes (1.0 - 1.4 mV) are evenly spaced with intervals of roughly 0.8 - 1.0 seconds, with smaller P and T waves filling out the space between R-peaks.

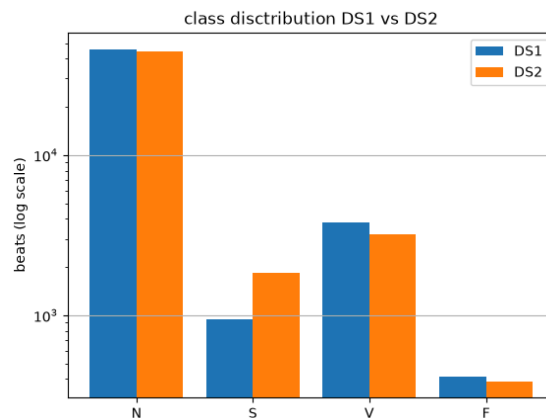
Example heartbeat from each class (DS1):



Selected representative segmented beats of each class are windowed around the R-peak.

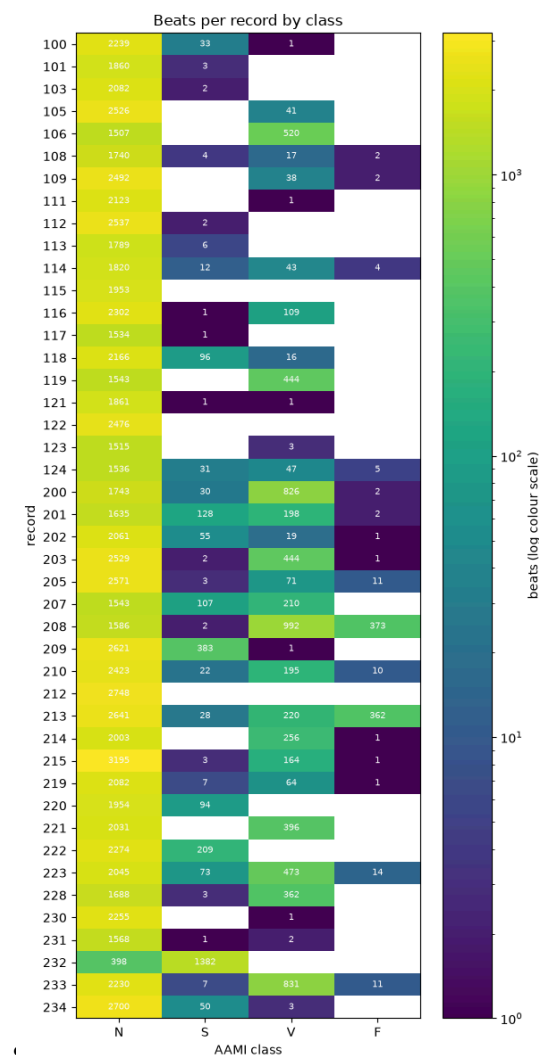
Represented class	Description
N (normal)	P wave, followed by a narrow and clean QRS, then a T wave.
S (supraventricular)	The QRS is similar to a normal beat (N); the distinguishing feature is timing (it comes early, with an abnormal P wave).
V (ventricular)	The QRS is wide and abnormal, with no preceding P wave and T wave is deflecting the QRS.
F (fusion)	The resulting QRS is a blend of a normal and a ventricular (V) beat.

Class distribution (DS1 vs DS2):



Beat counts per class were demonstrated on a log y-axis, and separated by dataset splits. N represents the most populated class, outnumbering the F class (the least populated) by roughly 110:1. As a result, a naive model would reach 90% accuracy while detecting no arrhythmia, making accuracy an inadequate metric. Explicit imbalance handling will be required. Also, a distribution difference is present between the dataset splits. The S class is roughly as twice as frequent in DS2, compared to DS1, which will be relevant during the model performance validation.

Record distribution by class:



The record distribution is presented in the form of a record by class heatmap of beat counts (cells annotated; blank = absent).

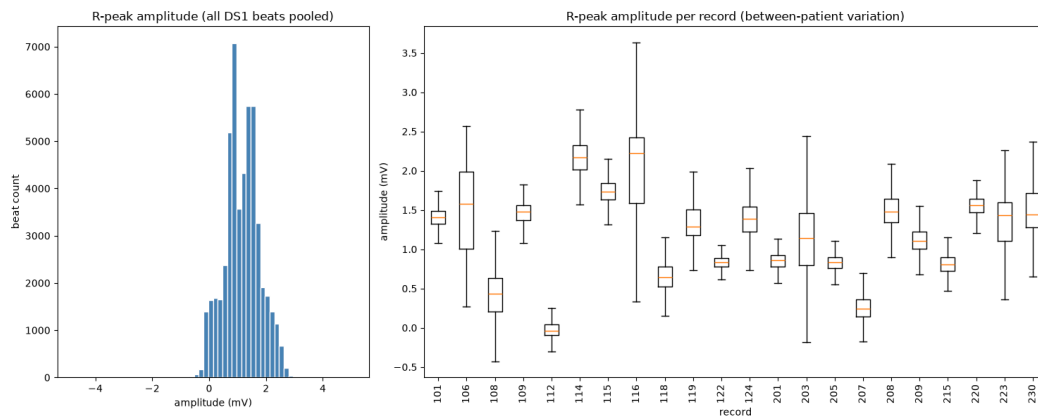
Class N: is present in all 44 records, with thousands of beats in nearly every one. This class can be considered the most well-represented across all patients. The one outlier is record 232, which holds 398 normal beats.

Class S: appears in most records but in the majority represents small number of beats per record (1-128 beats). Two main outliers are record 232 that contains 1,382 S beats and record 209 with 383 S beats, so a large share of the class comes from very few patients.

Class V: present in a majority of records and often with meaningful counts (e.g. 208 with 992 beats, 233 with 831 beats, 200 with 826 beats), but also absent in some records. It is the best-represented of the three abnormal classes.

Class F: appears in only 17 records, and in most of them often 1-5 beats. Almost the entire class is carried by records 208 (373 beats) and 213 (362 beats). F is effectively a two-patient class.

Signal amplitude distribution (DS1):

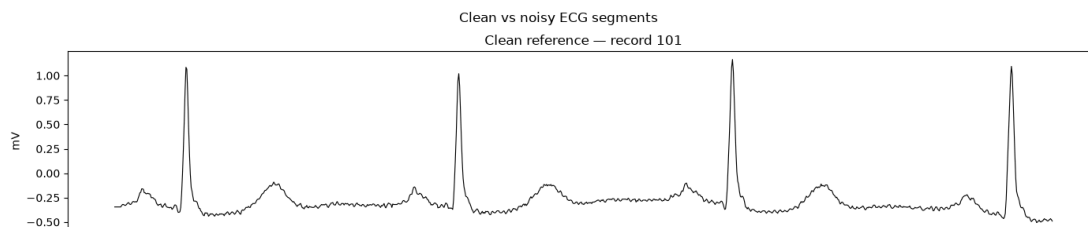


Left diagram: the histogram shows a single distribution of every DS1 beat. R-peak amplitude clusters around 1 mV, spanning roughly 0 - 3 mV, with one broad peak of 7000 beat counts at around 0.8 mV. This suggests a fairly uniform amplitude range across the dataset.

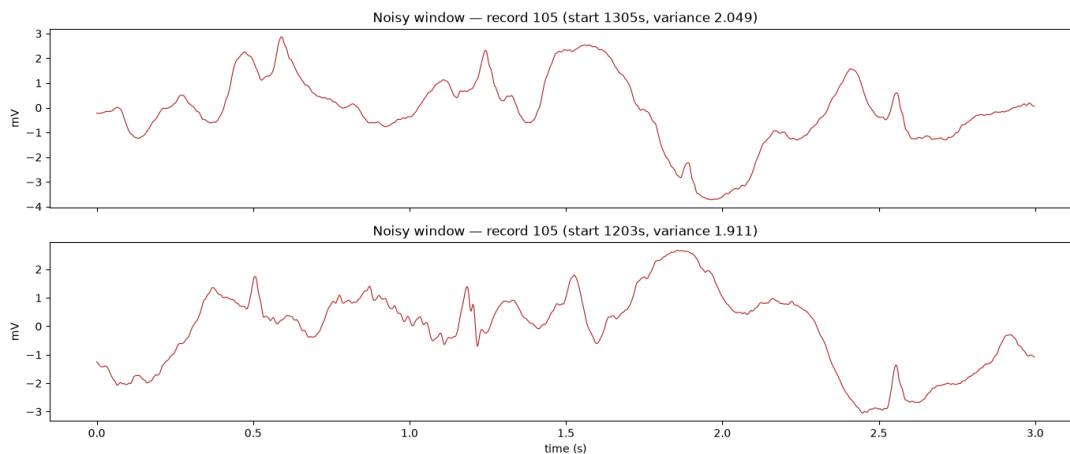
Right diagram: the box plot shows the same pooled distribution per record, which allows to show between-patient variation. Median R-peak height ranges from 0 mV (record 112) to above 2 mV (records 114, 116).

The amplitude per record visualisation reveals that the spread demonstrated in the histogram is driven largely by the between-patient differences, rather than by variability within any one recording, where amplitudes are comparatively tight.

Examples of noisy ECG signals:



Example of a 3-second window from record 101 with well-formed QRS complexes on a stable baseline to compare noisy ECG signals to.



Two segments of unstable, high-variance signal from record 105, dominated by large baseline wander and motion artefact, with amplitude swing from around -3 mV to around 2.5 mV.

2.4 Data Quality

Investigation.

Missing data: No missing samples in any of the 44 records, no gap handling is required.

Extreme amplitudes: Amplitudes range from -5.120 to 5.115 mV, the widest records are in 116 (8.27 mV range), 203 (4.84 mV) and 213 (4.7 mV). Except for 116, the min/max value stays near the typical range, showing the signal and not isolated spikes.

Corrupted segments: Record 116 has 928 samples at the 5.11 mV limit, the true peak is cut off. It's around 2.6 seconds of the recording and affects only 116 in the DS1. Clipped peaks can't be recovered, so it's a known defect.

Class imbalance: N has 89.48% of beats, V 6.96%, S 2.76% and F 0.8%, a classifier which predicts N for every beat will score 89.5% accuracy, while it would not detect arrhythmia at all. This is why accuracy alone is rejected as a success criterion, and why the balancing strategy is required

Differences between recordings: Per-record amplitude standard deviation is 0.165 mV (record 232) to 0.671 mV (record 213), the same beat appears several times bigger in one recording than another, which motivates per-record normalisation

Differences between ECG channels: The second lead is not the same in every record, and two leads in a record can have different amplitudes. In record 103, the V2 channel varies more than twice from MLII, while in record 105, MLII varies more than V1. If channels are picked by index, records would be read from the wrong lead without any errors, because of this, the lead is selected by name.

Possible noisy regions: Each record is split into 10-second windows, the disturbance ratio compares the noisiest window with the typical window. A value near 1 means that the record is clean, while a high value means that the part is noisier than other parts. The most disturbed records are 121, 108, 111, 116 and 207, while the best are 215, 221, 214, 231, and 209.

2.5 Leakage Analysis

Explain how data leakage could occur.

Data leakage is an important problem in this project because many heartbeats can come from the same subject. In this project, this could happen if individual heartbeats are randomly divided into training and test sets. Since many heartbeats from the same ECG recording or subject can have similar characteristics, the model may learn subject-specific patterns rather than general characteristics of the different heartbeat classes. As a result, the test set would not represent truly unseen subjects, and the measured performance could be misleadingly high.

To prevent this, the data is split by subject/ECG recording, rather than by individual heartbeat. All heartbeats from the same subject will be kept in the same dataset. The training set will therefore contain different subjects from the final test set.

This means that when the final model is evaluated, the test data will come from subjects that the model has not seen during training. This gives a more realistic measure of how well the model can classify heartbeats from previously unseen ECG recordings.

Even with a split by record, the test set dominates a few patients for the rare classes, f.ex record 232 has 1382 of the 1837 S beats in DS2 (75%), while record 213 holds 362 of the 388 F beats (93%). This means the score for S and F is close to a measurement on one patient, and a single recording can move the result a lot.

Records 201 and 202 are two parts of the same subject, 201 is in DS1 while 202 is in DS2. The model sees one part of this subject during training and one during testing, so the test set isn't subject independent. The same is inside DS1, the 22 development records are divided into 16 training and 6 validation records, no recording is used for both training and validation.

3. Data Preparation

3.1 Record Selection

List which records are used and what for.

Split	Records	N	S	V	F	Total
Train	16	34385 (90%)	731 (1.91%)	2710 (7.09%)	380 (1%)	38206
Validation	6	11467 (89.64%)	213 (1.66%)	1078 (8.43%)	34 (0.27%)	12792
DS1 Total	22	45866	944	3788	415	51013

The final test set is DS2 (22 records), which is not used during development. The DS1 (22 records) is divided into training and validation sets. The validation records are chosen so that the N, S, and V match the DS1 distribution, class F could not be balanced, since record 208 alone contains 373 of the 415 F beats.

3.2 ECG Channel Selection

Which ECG channel/lead is used and why.

All beats are taken from MLII, the second channel is not used. MLII is available in all 44 records, while the second channel is not everywhere. It is V1 in most records, and V2/V4/V5 in a few. Using MLII means that every record provides the same type of signal, this is important because the model is trained on one group and tested on the other. The usage of one channel also decreases the input size, which is important for the device with limited memory. If the lead is missing no replacement is needed. Only records 102 and 104 lack the MLII, both are excluded as paced records. The lead is selected by name and not by channel number, because record 114 stores MLII as the second channel instead of the first.

3.3 Beat Extraction

Describe how annotations were obtained.

The R-peak position of every beat is taken from the .atr annotation, they were placed by doctors, so no detection error enters the result. A deployed system would need its own detector and errors in it would shift the window. The input is a fixed size signal of shape (1, 200).

Samples before the annotated beat: 100 samples (0.278s at 360Hz).

Samples after the annotated beat: 100 samples (0.278s at 360Hz).

Final input length: 200 samples (0.556s) in a single channel, the window covers the P wave, the QRS complex and the start of the T wave, which are the parts that separate the classes. The T wave is not fully contained, as it is past the window size. At high heart rates, the window captures parts of the other beats, this is accepted because of the surrounding context carries the information. Beats which window would extend past the start or end of a recording can't be extracted and, because of this, are removed, 13 in the training set, 2 in the validation, and 15 in the test set, overall 30 beats out of 100718.

3.4 Signal Preprocessing

Augmentations used during training.

The operations are in this order: raw ECG -> filtering -> per-record normalisation -> clipping -> segmentation -> model input. All three signal operations are applied to the complete recording before its cut into beats.

Filtering (0.5 - 40 Hz): The low cutoff removes baseline wander, the slow drift caused by breathing and electrode movement, the high cutoff removes muscle activity, zero phase filtering runs the filter so that the beat is not shifted in time and the R-peak stays where the annotation was.

Normalisation: each recording is scaled to zero mean, section 2.4 shows that the amplitude standard deviation is from 0.165 to 0.671 mV. Without this step, the same beat would reach the model at very different size, because of the patient and equipment, and the model can learn the amplitude of a recording instead of the shape of the beat.

Clipping: After normalisation, values are limited to the range of ± 12 standard deviations, this removes isolated spikes, because they can dominate the input. The limit is high, so that the normal peaks are untouched, except for one in the test and one in the training sets out of 100718 .

Segmentation: The filtered, normalised signal is cut into fixed 200 sample windows, all records are at 360 Hz.

3.5 Label Preparation

Complete mapping.

The original MIT-BIH annotations are grouped into four final heartbeat classes:

Original MIT-BIH annotations	Final class	Description
N, L, R, e, j	N	Normal
A, a, J, S	S	Supraventricular ectopic
V, E	V	Ventricular
F	F	Fusion

In addition, several annotation types are excluded from the four-class classification task. The Q, /, and f annotations are not included because they are unclassifiable or too rare. Other annotations such as rhythm changes (+), signal quality changes (~), comments ("), non-conducted P waves (x), artefacts (!), and episode boundaries ([,]) are also excluded because they do not represent one of the four target heartbeat classes.

3.6 Class Imbalance

Report imbalance discovered.

The dataset is strongly imbalanced. Among the 100,718 beats used for the four-class classification task, N is the dominant class with 90,125 beats (89.5%), while S contains 2,781 beats (2.8%), V contains 7,009 beats (7.0%), and F contains only 803 beats (0.8%).

Class	Number of beats	% of total
N	90 125	89.5%
S	2 781	2.8%
V	7 009	7.0%
F	803	0.8%

This is a severe imbalance, N outnumbers F by more than 100:1. It is important because a model could obtain a high overall accuracy by mostly predicting the N class while performing poorly on the minority classes.

Chosen strategy: class weights + focal loss,

Class weights will be used as the primary approach. It will give a higher penalty to mistakes on the minority classes and a lower penalty to mistakes on the majority N class. This should encourage the model to pay more attention to S, V and especially F without artificially creating additional ECG signals.

Focal loss will be tested as an alternative to weighted cross-entropy. It reduces the contribution of easily classified examples and focuses more on difficult examples, which may be useful given the strong class imbalance in the dataset. The performance of focal loss and weighted cross-entropy will be compared on the validation set using record-independent evaluation.

3.7 Data Augmentation

Augmentations used during training.

Amplitude scaling (random factor): targets the between-patient amplitude variation (Section 2.4) and will encourage the model to rely on beat shape rather than absolute amplitude.

Small temporal shifts: the final system release will rely on its own R-peak detection rather than on pre-defined annotations, so the window focus shift will make the model robust to detect imprecisions.

Additive ECG noise: targets the 6 dB noise robustness success criteria (Section 1.4) and the artifact observed in Section 2.3. The evaluation of noise at test time is kept separate from the training noise model to avoid misleading robustness estimates.

Baseline wander: augmentations should account for the drift observed in the low frequencies, which are visible in the noisy sections of the recordings (Section 2.3). The high-pass preprocessing filter proposed in Section 3.4 should remove drift below 0.5 Hz. This augmentation will be applied before filtering so that it runs through the entire pipeline rather than being canceled out.

Augmentation must only be applied to the training data.

3.8 Final Dataset

Final dataset description.

The table reports the final beat counts after the preprocessing, excluding the 30 beats whose analysis window extended past the start or end of a recording (Section 3.3).

Split	Subjects/records	N	S	V	F	Total
Train	16	34,385	731	2,710	380	38,206
Validation	6	11,467	213	1,078	34	12,792
Test (DS2)	22	44,259	1,837	3,221	388	49,705

All beats use the single MLII lead (Section 3.2) with a fixed input shape. **Train** and **Validation** are drawn from the DS1 records, while **Test** is the DS2 set.

3.9 Reproducibility

Parameters to reproduce the analysis conducted.

- **Software versions:**
 - Python 3.14.7
 - Wfdb 4.3.1
 - Numpy 2.5.2
 - Scipy 1.18.1
 - Pandas 3.0.5
 - Matplotlib 3.11.0
- **Random seed:** 42, applied to record-level operations, augmentation, model initialization.
- **Preprocessing parameters:** band-pass filter 0.5-40 Hz, order 3, zero-phase, pre-record z-score normalisation (zero mean, unit variance).
- **Sampling frequency:** 360 Hz (no resampling).
- **Input size:** 100 samples before + 100 samples after the annotated R-peak, therefore 200 samples (0.556 s), single channel (MLII).

Appendix

AI Usage

AI was used during the project as a learning aid and consultation agent. It was used to explain domain concepts we were unfamiliar with, such ECG signals, the MLII lead, and the AAMI beat classes. It was also used to guide our implementation methods and how to approach certain aspects of the data analysis. Also was used to refine wording of the report. The AI **was not** used to generate any results or conclusions.

AI tools used: Claude, Grammarly.